

STAGE 5 INDUSTRIAL TECHNOLOGY ELECTIVE - METAL INDUSTRY

BBQ and Case | 45-period teaching program

Website-complete teacher review copy | Industrial Technology 7-10 Syllabus (2025)

Course	Stage 5 Industrial Technology elective - Metal industry	Delivery	2027 18 weeks
Timetable	45 x 60-minute periods	Pattern	5 periods per fortnight
Lesson balance	9 theory 9 project/folio 27 practical	Syllabus	2025 syllabus - approved early use
Status	Local teacher review only	Project	Produce the approved BBQ and sheet-metal carry case project
Website coverage: 27 named theory sections, 27 quiz/check sets (270 questions) and 27 scaffolded responses are scheduled. Practical lessons remain the majority.			

Industrial Technology outcomes

Outcome	Official wording
INT5-SAF-01	applies risk management and safe work practices in industrial technology contexts
INT5-COM-01	communicates ideas and processes in industrial technology contexts
INT5-ENV-01	assesses the impact of industrial activities and practices on society and the environment
INT5-DES-01	selects and applies design, project management and technical skills in project production
INT5-USE-01	selects and justifies the use of materials, tools and equipment to undertake projects
INT5-EVL-01	evaluates the functional, aesthetic, economic and environmental qualities of products
INT5-INP-01	analyses and applies a range of industrial processes in production contexts
INT5-PRO-01	demonstrates practical skills in the production of projects
INT5-IND-01	investigates and explains a range of industry careers, enterprise opportunities and practices

Teacher to confirm before use

When	Confirm	Owner
Before Period 1	Assessment dates, submission method, class supports and adjustments	Teacher / faculty
Before practical work	Current plan revision, permitted variations, induction, SOPs, tools, PPE, supervision and risk controls	Teacher / faculty
Before finishing	Approved finish, SDS/product instructions, area, drying and storage arrangements	Teacher / faculty

Teaching program | Periods 1-10

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
1	Site/theory	Module 1: website	Read and discuss: Reading the BBQ-and-case brief as a production contract; Reading the BBQ-and-case brief as a production contract knowledge check (10 questions); written response. Save or show the completed website evidence.	Design factors and project specifications	Website evidence checked INT5-DES-01, INT5-COM-01
2	Project/folio	Module 1: folio	Read and complete: From hazard review to a task-specific WMS; From hazard review to a task-specific WMS knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Risk management and risk-reduction procedures	Folio + website evidence INT5-SAF-01
3	Practical	Module 1: site + practical	First 20 min: Inputs, processes and outputs as a planning model; Inputs, processes and outputs as a planning model knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Accurately measure and mark out metal	Website + practical check INT5-USE-01, INT5-PRO-01
4	Practical	Module 1: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Metal properties and justified material selection	Practical quality check INT5-USE-01, INT5-ENV-01
5	Practical	Module 1: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Metal properties and justified material selection	Photo + quality note INT5-USE-01, INT5-ENV-01
6	Site/theory	Module 2: website	Read and discuss: Selecting stock form by function and manufacture; Selecting stock form by function and manufacture knowledge check (10 questions); written response. Save or show the completed website evidence.	Metal properties and justified material selection	Website evidence checked INT5-USE-01, INT5-ENV-01
7	Project/folio	Module 2: folio	Read and complete: Turning drawings into freehand sketches and a spreadsheet cutting list; Turning drawings into freehand sketches and a spreadsheet cutting list knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Accurately measure and mark out metal	Folio + website evidence INT5-USE-01, INT5-PRO-01
8	Practical	Module 2: site + practical	First 20 min: Datums, orientation and quality gates for rails and end caps; Datums, orientation and quality gates for rails and end caps knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Accurately measure and mark out metal	Website + practical check INT5-USE-01, INT5-PRO-01
9	Practical	Module 2: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Accurately measure and mark out metal	Practical quality check INT5-USE-01, INT5-PRO-01
10	Practical	Module 2: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Accurately measure and mark out metal	Photo + quality note INT5-USE-01, INT5-PRO-01

Teaching program | Periods 11-20

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
11	Site/theory	Module 3: website	Read and discuss: Hole location is a design feature, not just a mark; Hole location is a design feature, not just a mark knowledge check (10 questions); written response. Save or show the completed website evidence.	Accurately measure and mark out metal	Website evidence checked INT5-USE-01, INT5-PRO-01
12	Project/folio	Module 3: folio	Read and complete: Drill-press principles behind the teacher-issued SOP; Drill-press principles behind the teacher-issued SOP knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Risk management and risk-reduction procedures	Folio + website evidence INT5-SAF-01, INT5-INP-01
13	Practical	Module 3: site + practical	First 20 min: Inspecting drilled features and responding to faults; Inspecting drilled features and responding to faults knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Use machining and fabrication tools and equipment	Website + practical check INT5-SAF-01, INT5-INP-01
14	Practical	Module 3: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Metal joining and fabrication techniques	Practical quality check INT5-INP-01, INT5-PRO-01
15	Practical	Module 3: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Metal joining and fabrication techniques	Photo + quality note INT5-INP-01, INT5-PRO-01
16	Site/theory	Module 4: website	Read and discuss: Reading a lathe operation through the WMS; Reading a lathe operation through the WMS knowledge check (10 questions); written response. Save or show the completed website evidence.	Metal lathe operations	Website evidence checked INT5-SAF-01, INT5-INP-01
17	Project/folio	Module 4: folio	Read and complete: Cutting action, reference surfaces and dimensional control; Cutting action, reference surfaces and dimensional control knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Accurately measure and mark out metal	Folio + website evidence INT5-USE-01, INT5-PRO-01
18	Practical	Module 4: site + practical	First 20 min: Surface quality, fit and documented teacher review; Surface quality, fit and documented teacher review knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Prepare metal surfaces for finishes	Website + practical check INT5-INP-01, INT5-PRO-01
19	Practical	Module 4: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Technical skill for quality projects	Practical quality check INT5-PRO-01, INT5-INP-01
20	Practical	Module 4: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Technical skill for quality projects	Photo + quality note INT5-PRO-01, INT5-INP-01

Teaching program | Periods 21-30

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
21	Site/theory	Module 5: website	Read and discuss: Planning plate work from verified references; Planning plate work from verified references knowledge check (10 questions); written response. Save or show the completed website evidence.	Accurately measure and mark out metal	Website evidence checked INT5-USE-01, INT5-PRO-01
22	Project/folio	Module 5: folio	Read and complete: Grills and guard: function, spacing and manufacturability; Grills and guard: function, spacing and manufacturability knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Quality checks for metals and components	Folio + website evidence INT5-EVL-01, INT5-PRO-01
23	Practical	Module 5: site + practical	First 20 min: Fit-up before permanent joining; Fit-up before permanent joining knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Metal joining and fabrication techniques	Website + practical check INT5-INP-01, INT5-PRO-01
24	Practical	Module 5: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Document, reflect on and evaluate project development	Practical quality check INT5-COM-01, INT5-DES-01
25	Practical	Module 5: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Document, reflect on and evaluate project development	Photo + quality note INT5-COM-01, INT5-DES-01
26	Site/theory	Module 6: website	Read and discuss: Stability begins with geometry and matched components; Stability begins with geometry and matched components knowledge check (10 questions); written response. Save or show the completed website evidence.	Industrial Technology terminology	Website evidence checked INT5-COM-01
27	Project/folio	Module 6: folio	Read and complete: Assembly order, access and hold points; Assembly order, access and hold points knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Document, reflect on and evaluate project development	Folio + website evidence INT5-COM-01, INT5-DES-01
28	Practical	Module 6: site + practical	First 20 min: Quality assurance during structural assembly; Quality assurance during structural assembly knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Quality checks for metals and components	Website + practical check INT5-EVL-01, INT5-PRO-01
29	Practical	Module 6: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Sustainable practices in the metal industry	Practical quality check INT5-ENV-01, INT5-IND-01
30	Practical	Module 6: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Sustainable practices in the metal industry	Photo + quality note INT5-ENV-01, INT5-IND-01

Teaching program | Periods 31-40

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
31	Site/theory	Module 7: website	Read and discuss: Surface condition controls finish performance; Surface condition controls finish performance knowledge check (10 questions); written response. Save or show the completed website evidence.	Prepare metal surfaces for finishes	Website evidence checked INT5-INP-01, INT5-PRO-01
32	Project/folio	Module 7: folio	Read and complete: Selecting and applying a finish from authoritative information; Selecting and applying a finish from authoritative information knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Prepare metal surfaces for finishes	Folio + website evidence INT5-INP-01, INT5-PRO-01
33	Practical	Module 7: site + practical	First 20 min: Finish inspection, maintenance and environmental responsibility; Finish inspection, maintenance and environmental responsibility knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Sustainable practices in the metal industry	Website + practical check INT5-ENV-01, INT5-IND-01
34	Practical	Module 7: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Prepare metal surfaces for finishes	Practical quality check INT5-INP-01, INT5-PRO-01
35	Practical	Module 7: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Prepare metal surfaces for finishes	Photo + quality note INT5-INP-01, INT5-PRO-01
36	Site/theory	Module 8: website	Read and discuss: From three-dimensional case to flat development; From three-dimensional case to flat development knowledge check (10 questions); written response. Save or show the completed website evidence.	Accurately measure and mark out metal	Website evidence checked INT5-USE-01, INT5-PRO-01
37	Project/folio	Module 8: folio	Read and complete: Bending sequence, accumulated error and case fit; Bending sequence, accumulated error and case fit knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Cut, shape and form sheet metal and metal sections	Folio + website evidence INT5-INP-01, INT5-PRO-01
38	Practical	Module 8: site + practical	First 20 min: Hinges and handle as an evidence-based design problem; Hinges and handle as an evidence-based design problem knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Design factors and project specifications	Website + practical check INT5-DES-01, INT5-COM-01
39	Practical	Module 8: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Quality checks for metals and components	Practical quality check INT5-EVL-01, INT5-PRO-01
40	Practical	Module 8: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Quality checks for metals and components	Photo + quality note INT5-EVL-01, INT5-PRO-01

Teaching program | Periods 41-45

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
41	Site/theory	Module 9: website	Read and discuss: PMI evaluation that moves beyond opinion; PMI evaluation that moves beyond opinion knowledge check (10 questions); written response. Save or show the completed website evidence.	Quality checks for metals and components	Website evidence checked INT5-EVL-01, INT5-PRO-01
42	Project/folio	Module 9: folio	Read and complete: Building the project report as an evidence chain; Building the project report as an evidence chain knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Document, reflect on and evaluate project development	Folio + website evidence INT5-COM-01, INT5-DES-01
43	Practical	Module 9: site + practical	First 20 min: Final quality assurance and authentic submission; Final quality assurance and authentic submission knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Quality checks for metals and components	Website + practical check INT5-EVL-01, INT5-PRO-01
44	Practical	Module 9: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Document, reflect on and evaluate project development	Practical quality check INT5-COM-01, INT5-DES-01
45	Practical	Module 9: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Document, reflect on and evaluate project development	Photo + quality note INT5-COM-01, INT5-DES-01

Evidence, feedback and practical controls

Check	Look for	Respond
Website	Named theory, checks and responses completed	Revisit feedback and correct missed ideas
Folio	Decisions and process evidence tied to the project	Set one clear next step
Practical	Safe method, accuracy, quality and clean-down	Pause, demonstrate, practise and recheck

Use the current approved plan, local induction, SOPs, PPE, supervision, risk controls and product instructions for every practical lesson.